

User App Frontend

Buyer-facing ecommerce frontend for browsing, account access, wishlist, cart, checkout, orders, and profile management.

PREPARED DATE

July 6, 2026

DOCUMENT VERSION

1.0

PROJECT LOCATION

`frontend/user-app`

LOCAL URL

`http://localhost:3000`

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This manual covers only functionality found in the current code, routes, and API integrations. No fake credentials, fake products, or fake successful workflows were added.

1. Service Overview

The User App Frontend is a React, TypeScript, Vite buyer application. It provides public product browsing and protected buyer flows such as wishlist, cart, checkout, orders, addresses, profile, and notification preferences.

Area	Status	What The User Can Do
Home/catalog	Available	Browse published products and category chips. The current local stack returned 0 published products during verification.
Categories	Available	Open category cards and category-specific product result pages.
Search and filters	Available	Search by keyword, filter by category, brand, seller, price, rating, and stock, and sort results.
Product details	Available	Open product detail pages, view image gallery, variants, price, stock, seller, category, and product status when a product exists.
Authentication	Available	Register, verify OTP, login, logout, start forgot-password flow, and reset password.
Wishlist	Available	Add/remove products, view saved products, and move saved items to cart after login.
Cart	Available	Add products, change quantity, remove items, preview coupons, and proceed to checkout after login.
Checkout	Available	Select saved address, choose payment provider, review order, and create checkout through Order Service.
Payment provider widget	Partial	The UI shows payment intent details and a provider mount placeholder. Redirect works only if the backend returns a payment URL. Embedded Stripe/Razorpay SDK UI is not implemented in this frontend.
Orders	Available	List orders, view details, status timeline, delivery address, payment/fulfillment statuses, and cancel eligible orders.
Notification center	Partial	Notification preferences are implemented. An in-app notification inbox/list is not present.
Recommendations UI	Not Implemented	Recommendation gRPC client code exists, but no visible route or component uses it in the current UI.

Verified Local State

Frontend health	<code>http://localhost:3000/healthz</code> returned HTTP 200.
Categories API	<code>/api/v1/categories</code> returned category records.
Published products	<code>/api/v1/products?page=1&page_size=4&status=published</code> returned 0 products.

Search service	Search UI loaded. A direct keyword API check previously produced a transient gateway timeout, so search troubleshooting is included.
Buyer credentials	No seeded buyer/test user credentials were found in local docs, migrations, or runbooks.

2. Local Access And Dependencies

How To Run Locally

Method	Command	URL
Docker local stack	<code>docker compose up -d --build user-app</code>	<code>http://localhost:3000</code>
Manual Vite dev server	<code>cd frontend</code> <code>corepack pnpm dev:user</code>	Vite usually prints <code>http://localhost:5173</code> .
Health check	Open <code>http://localhost:3000/healthz</code>	Expected HTTP 200 in Docker mode.

Required Environment Variables

Variable	Default / Example	Purpose
<code>VITE_API_BASE_URL</code>	<code>http://localhost:8080</code>	REST API Gateway base URL.
<code>VITE_GRPC_WEB_BASE_URL</code>	<code>http://localhost:8099</code>	gRPC-Web Gateway base URL.
<code>VITE_GRPC_WEB_TIMEOUT_MS</code>	<code>5000</code>	gRPC-Web timeout in milliseconds.
<code>VITE_APP_ENV</code>	<code>local</code>	Controls local-only notices such as Mailpit OTP guidance.
<code>VITE_PAYMENT_PROVIDERS</code>	<code>stripe,razorpay</code>	Controls visible payment method radio options.

Required Backend/API Dependencies

Frontend Function	Backend/API Dependency	Endpoint Examples
Signup, login, OTP, password reset, logout	API Gateway, Auth Service, User Service, Notification Service for OTP delivery	<code>/api/v1/auth/signup</code> , <code>/api/v1/auth/login</code> , <code>/api/v1/auth/otp/send</code> , <code>/api/v1/auth/otp/verify</code>
Product browsing and details	API Gateway, Product Service, Search Service for search-only filters	<code>/api/v1/products</code> , <code>/api/v1/products/{id}</code> , <code>/api/v1/categories</code> , <code>/api/v1/search</code>
Autocomplete	API Gateway, Search Service	<code>/api/v1/search/autocomplete</code>
Wishlist	API Gateway, Wishlist Service, Product Service hydration	<code>/api/v1/wishlist</code> , <code>/api/v1/wishlist/items</code> , <code>/api/v1/wishlist/items/{product_id}/move-to-cart</code>
Cart and coupons	API Gateway, Cart Service, Product Service inventory	<code>/api/v1/cart</code> , <code>/api/v1/cart/items</code> , <code>/api/v1/cart/coupons/preview</code>
Checkout and payment	API Gateway, Cart Service, User Service addresses, Order	<code>/api/v1/orders/checkout</code> , <code>/api/v1/payments/{payment_id}/retry</code>

Frontend Function	Backend/API Dependency	Endpoint Examples
	Service, Product Service inventory, Payment Service	
Profile, addresses, notifications	API Gateway, User Service, Notification Service preferences	/api/v1/me , /api/v1/me/addresses , /api/v1/me/notification-preferences
Orders	API Gateway, Order Service, Payment Service status	/api/v1/orders , /api/v1/orders/{id} , /api/v1/orders/{id}/cancel

3. Application Map

Route	Access	Purpose
/	Public	Home page with category chips and latest published products.
/categories	Public	Category grid.
/category/:categoryId	Public	Category-specific product results with filters and sort.
/deals	Public	Popular products using <code>sort=popular</code> .
/search	Public	Search results with filters, facets, sort, and pagination.
/products/:productId	Public	Product details, variants, add to cart, wishlist.
/signup, /login	Public	Account creation and login.
/auth/otp	Public	OTP verification after signup or resend.
/forgot-password, /reset-password	Public	Password reset flow.
/wishlist, /cart, /checkout	Protected	Saved products, cart, and checkout.
/checkout/success, /checkout/failure	Protected	Payment result and retry screens.
/account	Protected	Account overview.
/account/profile, /account/addresses, /account/orders, /account/notifications	Protected	Profile, delivery addresses, orders, and notification preferences.

Main Header And Public Navigation

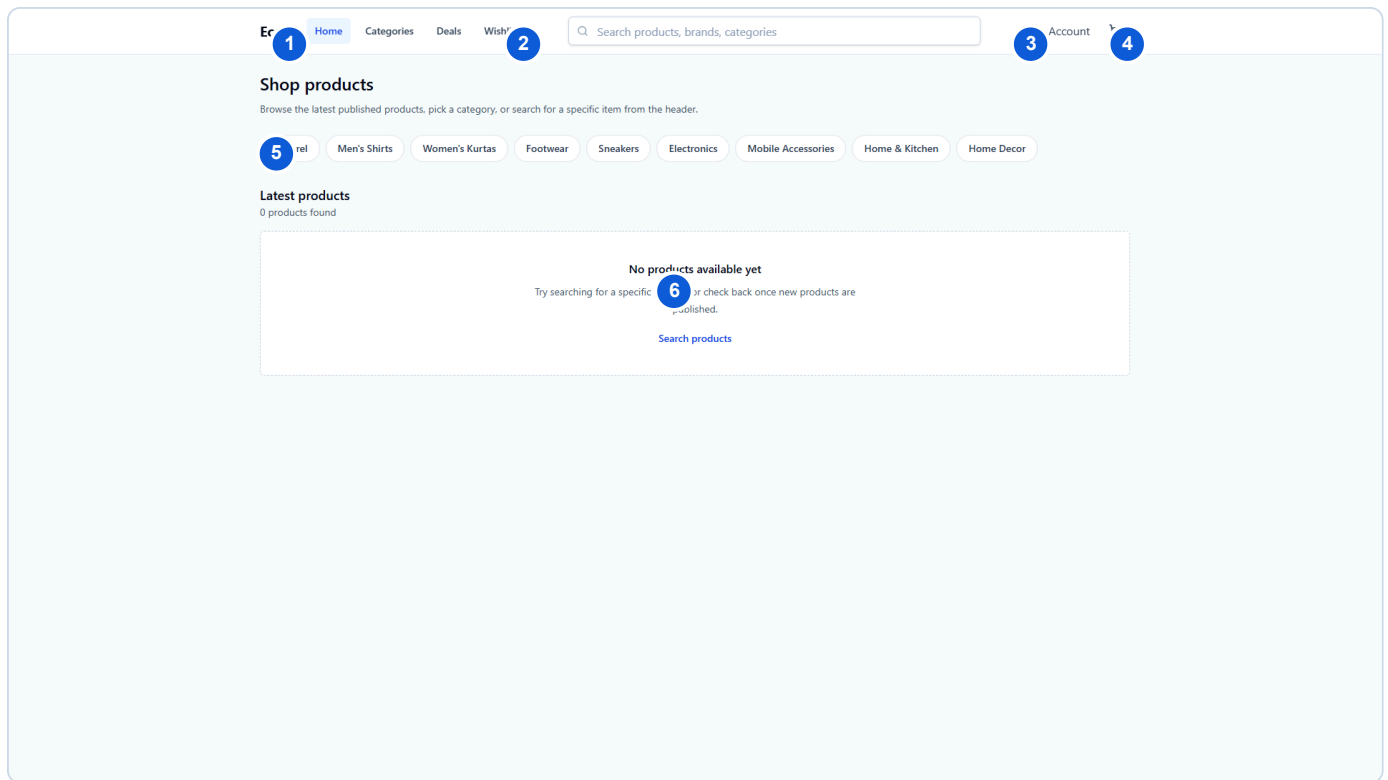


Figure 1: Home page, captured from the live local UI. Current local catalog has no published products.

1	Ecom logo: returns to the home page.
2	Search box: enter product, brand, or category text. Type 2 or more characters to request autocomplete suggestions.
3	Account menu: opens Login/Create account when signed out, or Profile/Addresses/Orders/Notifications/Logout when signed in.
4	Cart icon: opens cart. The badge shows total cart quantity after login/session detection.
5	Category chips: open category result pages.
6	Empty state: appears when no published products are returned.

4. New User Step-By-Step Guide

Open The Application

- 1 Start Docker local mode or the Vite dev server.
- 2 Open `http://localhost:3000` in a browser.
- 3 Confirm the header shows the Ecom logo, Home, Categories, Deals, Wishlist, search box, Account icon, and Cart icon.

Register New Account

The screenshot shows the 'Create your account' form in the Ecom application. The form is centered on a light blue background. It includes fields for Full name, Email, Phone, Password, and Confirm password. A password strength indicator is shown below the password field, listing requirements: at least 8 characters, one uppercase letter, one lowercase letter, and one number. A checkbox for terms and privacy policy is at the bottom. A blue 'Create account' button and a 'Login' link are at the bottom. Numbered callouts 1 through 7 highlight specific elements: 1. Full name field, 2. Email field, 3. Phone field, 4. Password field, 5. Password strength requirements, 6. Terms and privacy policy checkbox, 7. Login link.

Figure 2: Signup form. Password visibility is controlled by the eye icon.

- 1 Open Account -> Create account, or go to `/signup`.
- 2 Enter Full name, Email, optional Phone, Password, and Confirm password.
- 3 Use the password strength panel to satisfy all requirements: at least 8 characters, one uppercase letter, one lowercase letter, and one number.
- 4 Select the terms and privacy policy checkbox.
- 5 Select Create account.
- 6 If signup succeeds, the app sends an OTP and opens the OTP verification page.

Local email OTPs are captured in Mailpit at `http://localhost:8025` when `VITE_APP_ENV=local`. Local phone OTP delivery may be unavailable; use an email address for local testing.

Verify OTP

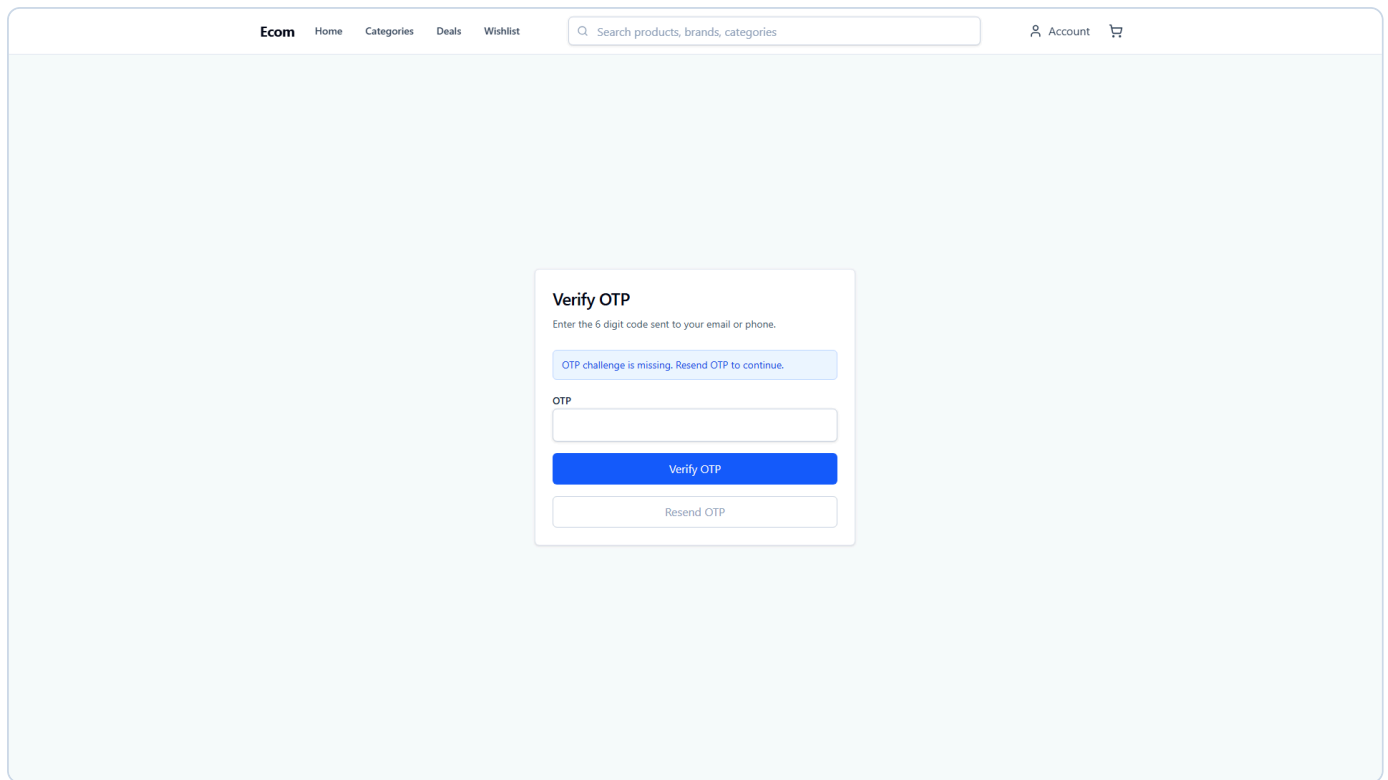


Figure 3: OTP page without a challenge. A normal signup flow passes the challenge id in the URL.

- 1 Open Mailpit and copy the 6 digit OTP from the email message.
- 2 Enter the OTP in the OTP field.
- 3 Select Verify OTP.
- 4 If needed, select Resend OTP. The button uses a 30 second cooldown after a successful resend.
- 5 After verification, the app redirects to Login with the notice "OTP verified. Please login to continue."

Login

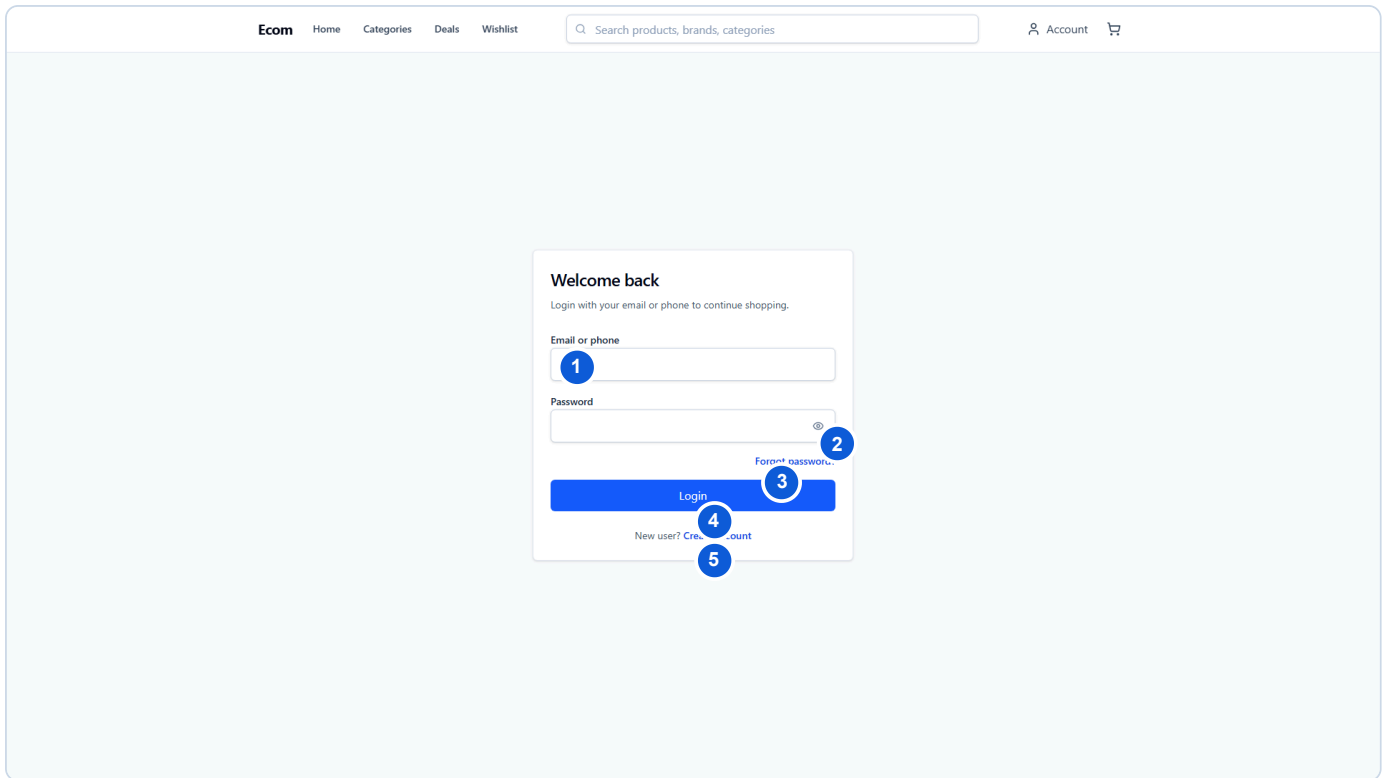


Figure 4: Login page.

- 1 Open Account -> Login, or go to `/login`.
- 2 Enter Email or phone.
- 3 Enter Password. Use the eye icon to show/hide the password.
- 4 Select Login.
- 5 If the user opened a protected page first, the app returns the user to that page after login through the `redirect` query parameter.

Browse Home Page/Dashboard

- 1 Select Home or the Ecom logo.
- 2 Select a category chip to open a category page.
- 3 If products are present, select a product image or title to open product details.
- 4 If no products are present, use Search products or wait until published products exist in Product Service.

Search

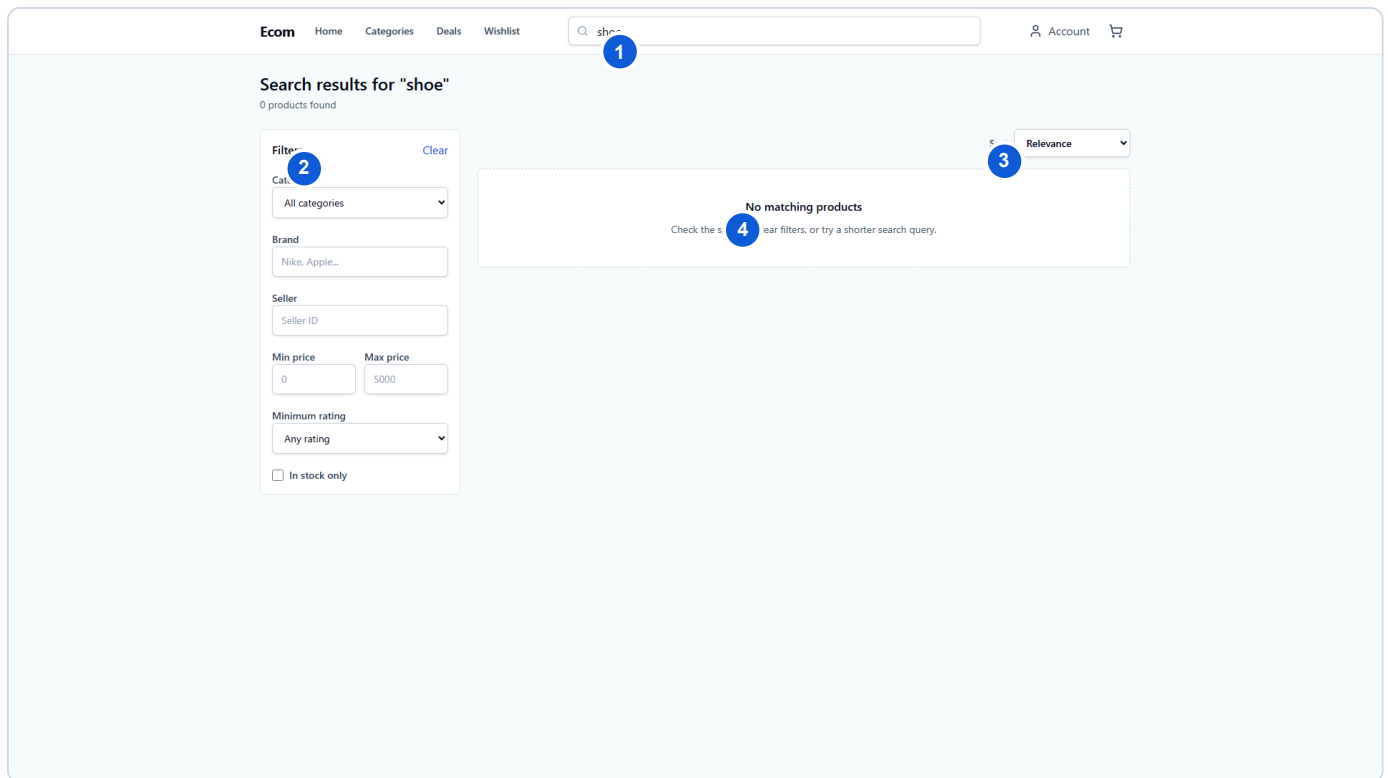


Figure 5: Search result page with filters. Current local data returned no matching products.

- 1 Type a keyword in the header search box.
- 2 Press Enter. Empty searches are ignored.
- 3 Use filters for category, brand, seller, min price, max price, minimum rating, and in-stock-only.
- 4 Use Sort to choose Relevance, Popular, Newest, Price Low to High, Price High to Low, or Top Rated.
- 5 Select Clear to reset filters while keeping the search query.

Category/Filter Usage

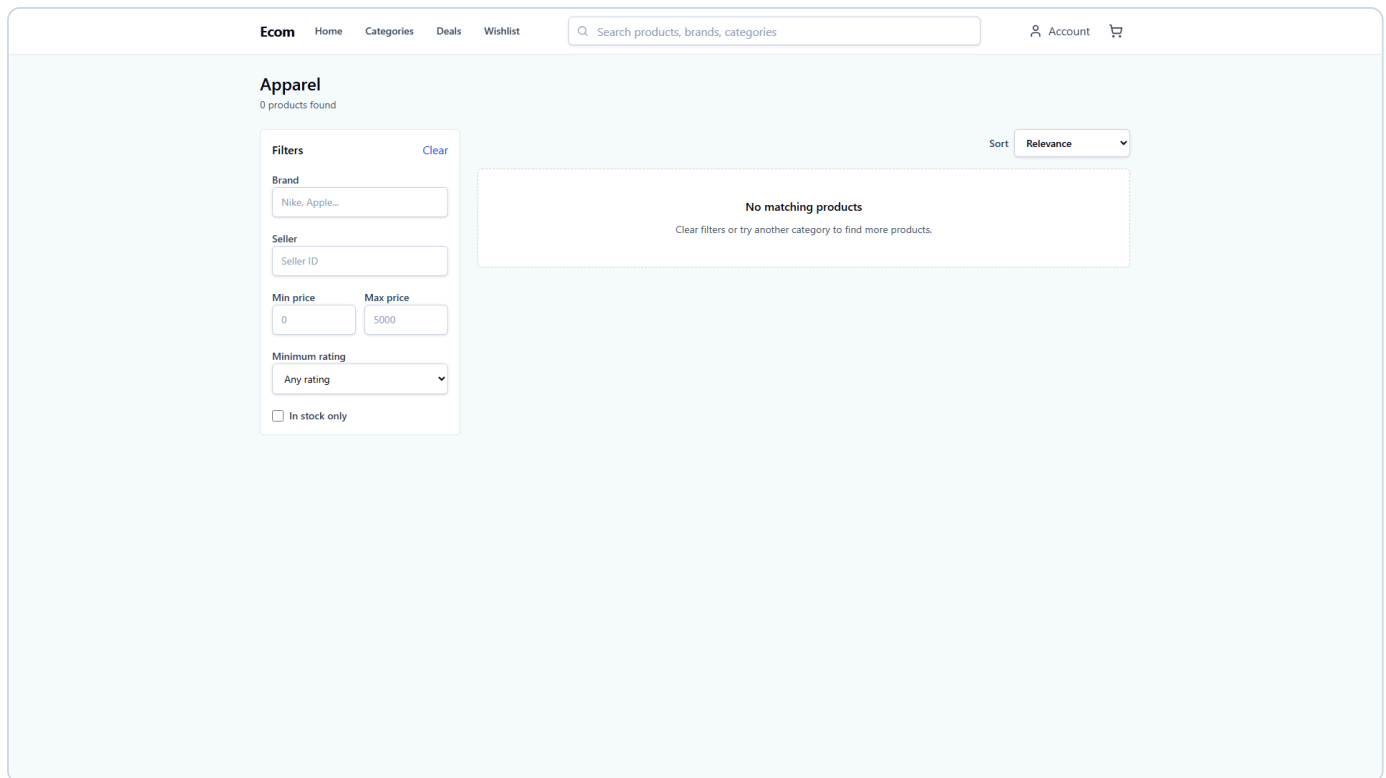


Figure 6: Category page. Category filter is hidden because the category is already selected.

- 1 Open Categories.
- 2 Select a category card, for example Apparel.
- 3 Use Brand, Seller, Min price, Max price, Minimum rating, and In stock only filters.
- 4 Use Clear to remove filters.

Product/Service Listing And Details

- 1 Listings show product image, brand, title, price, stock label, rating, wishlist heart, and Add button when products exist.
- 2 Select the product image/title to open details.
- 3 The detail page shows gallery, brand, title, price, description, variants, stock, Add to cart, Wishlist, Product ID, Seller, Category, Status, and Selected SKU.

Screenshot to be added: Product Listing With Published Products. Current local Product Service returned 0 published products.

Screenshot to be added: Product Detail Page With Real Product. A real published product id is required.

Wishlist

- 1 Select the heart icon on a listing/detail page.
- 2 If not logged in, the app redirects to Login.
- 3 After login, open Wishlist to review saved products.
- 4 Select Move to cart for available products, or use the trash icon to remove an item.

Screenshot to be added: Logged-in Wishlist Page. No seeded buyer user was found.

Cart

- 1 Select Add or Add to cart from a product listing/detail page.
- 2 If not logged in, the app redirects to Login.
- 3 Open Cart from the cart icon.
- 4 Use minus/plus to change quantity. Use Remove to remove the item.
- 5 Enter a coupon code and select Apply to preview the discount.
- 6 Select Proceed to checkout when the cart has items and no out-of-stock items.

Screenshot to be added: Logged-in Cart With Items. A logged-in buyer and product data are required.

Checkout And Payment

- 1 Open Checkout from the cart summary.
- 2 Select a saved delivery address. If no address exists, add one from Account -> Addresses first.
- 3 Select Card / Stripe or UPI / Razorpay based on configured providers.
- 4 Review order lines, subtotal, discount, and total.
- 5 Select Create payment.
- 6 If the backend returns a payment intent, the app displays Payment ready.
- 7 If a provider redirect URL is returned, select Continue to payment.

Payment provider UI is partial. The frontend contains a mount placeholder for secure provider SDK UI. It does not implement embedded Stripe/Razorpay SDK forms.

Screenshot to be added: Checkout Review Page. Requires logged-in buyer, cart item, and saved address.

Screenshot to be added: Payment Ready Page. Requires successful checkout response with payment intent.

Profile, Notifications, And Logout

- 1 Open Account -> Profile to update full name, phone, or avatar URL. Email is read-only.
- 2 Open Account -> Addresses to add, edit, delete, or mark a default address.
- 3 Open Account -> Orders to view order history and order details.
- 4 Open Account -> Notifications to save email, SMS, push, and marketing preferences.
- 5 Open Account -> Logout to clear the local session and cached buyer data.

Screenshot to be added: Logged-in Account/Profile/Addresses/Orders/Notifications Pages. No seeded buyer user was found.

5. Existing Or Test User Guide

No seeded buyer/test user credentials were found in `LOCAL_ACCESS_GUIDE.md`, runbooks, migrations, frontend code, or the local access matrix. This manual therefore does not provide login credentials.

Example Registration Data Format

Use your own local-only buyer data. Do not use production credentials in a local or shared environment.

Field	Expected Format	Rules
Full name	Buyer Name	2 to 120 characters.
Email	local-buyer@example.test	Required valid email address.
Phone	+15551234567	Optional. Local phone OTP may be disabled.
Password	Local-only password	8 to 128 characters, uppercase, lowercase, and number.
Confirm password	Same as password	Must match password.

Forgot And Reset Password

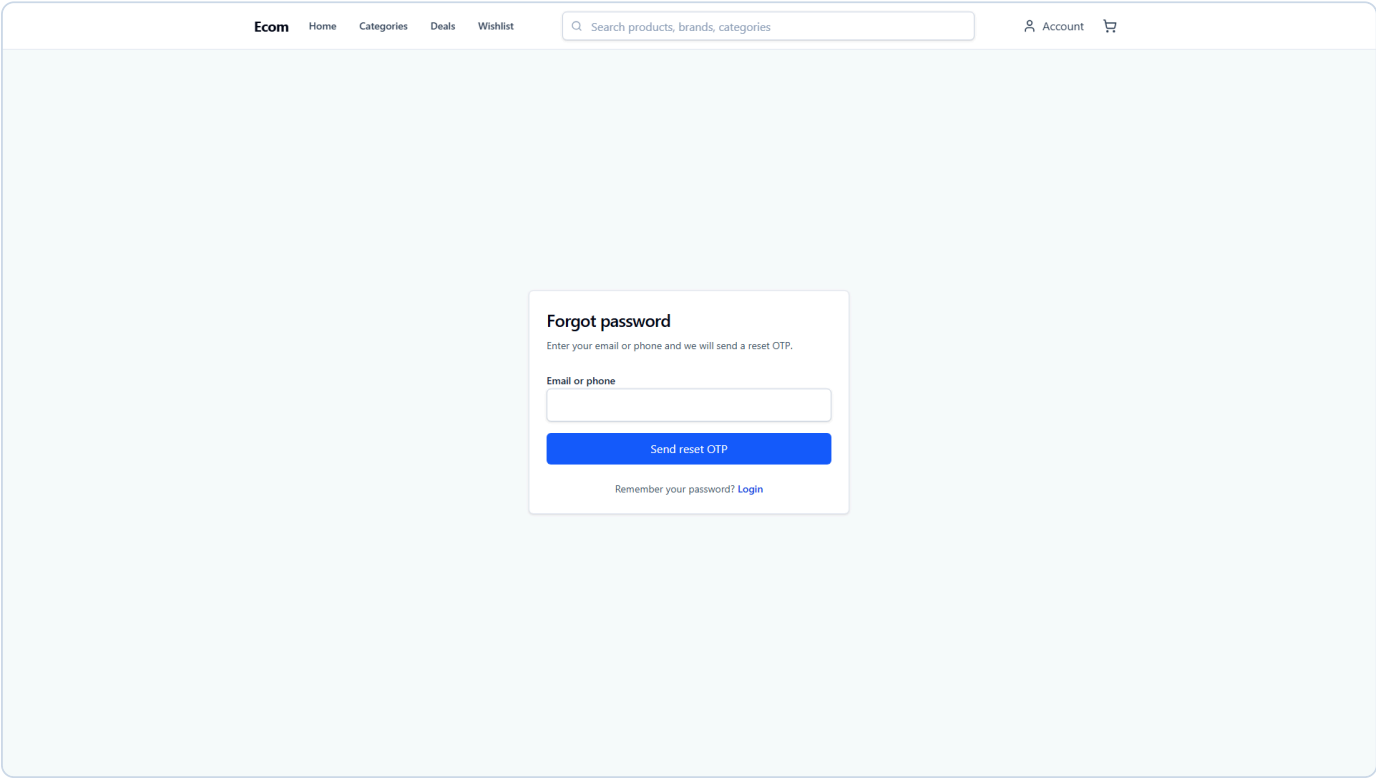


Figure 7: Forgot password page.

- 1 Open Login -> Forgot password.
- 2 Enter email or phone.
- 3 Select Send reset OTP.
- 4 Open reset link/page with the challenge id.

5 Enter OTP, new password, and confirm password.

6 Select Reset password.

6. Page And Control Reference

Shared Header, Menus, And Icons

Control	Available State	User Action
Ecom brand link	All pages	Opens Home.
Home	All pages	Opens latest published products.
Categories	All pages	Opens category grid.
Deals	All pages	Opens popular products view.
Wishlist	All pages	Opens protected wishlist. Signed-out users are redirected to Login.
Search box and search icon	All pages	Typing updates suggestions after 2 characters. Enter opens Search results.
Account icon	All pages	Opens account menu. Signed out: Login/Create account. Signed in: Profile, Addresses, Orders, Notifications, Logout.
Cart icon and badge	All pages	Opens Cart. Badge shows total quantity and caps display at 99+.
Browse menu	Mobile width only	Opens/collapses Home, Categories, Deals, Wishlist links. Icon changes between menu and close.

Authentication Forms

Page	Fields And Buttons	Validation/Status Messages
Signup	Full name, Email, Phone, Password, Confirm password, terms checkbox, Create account, Login link, password eye buttons.	Email is required, valid email required, full name min 2/max 120, phone valid/max 24, password min/max and uppercase/lowercase/number, confirm required, passwords must match, terms required.
OTP	OTP input, Verify OTP, Resend OTP.	OTP challenge is missing, Enter the 6 digit OTP, local Mailpit notice, resend success/failure, phone OTP unavailable in local stack.
Login	Email or phone, Password, Forgot password link, Login button, Create account link, password eye button.	Email or phone required/too long/invalid, password required, Login failed, OTP verified notice, password reset notice.
Forgot password	Email or phone, Send reset OTP, Login link.	Email or phone required/too long/invalid, could not start password reset.
Reset password	OTP, New password, Confirm new password, password	Missing challenge, OTP format, password rules, confirm required, passwords must match, resend success/failure.

Page	Fields And Buttons	Validation/Status Messages
	strength, Reset password, Resend OTP, Login link.	

Browse/Search/Product Controls

Control	Page	Behavior
Category chips/cards	Home, Categories	Open <code>/category/:categoryId</code> .
Product image/title	Listings	Open <code>/products/:productId</code> .
Heart icon	Listings/detail	Add/remove wishlist item. Redirects to Login if unauthorized.
Add / Add to cart	Listings/detail	Adds selected product variant to cart. Disabled if no variant or out of stock.
View cart	Product detail after add	Appears after successful add and opens Cart.
Sort select	Search/category	Supports Relevance, Popular, Newest, Price: Low to High, Price: High to Low, Top Rated.
Filter fields	Search/category	Category, Brand, Seller, Min price, Max price, Minimum rating, In stock only. Category filter is hidden on category-specific page.
Pagination	Home/search/category/deals	Previous/Next when total pages are greater than one.
Variant radio cards	Product detail	Select SKU/variant and shows attributes, price, and stock. If no variants exist, the UI says variant information is unavailable.

Cart And Wishlist Controls

Control	Behavior	Messages
Quantity minus	Decreases quantity by 1. Disabled at quantity 1 or out of stock.	Quantity could not be updated.
Quantity plus	Increases quantity by 1.	Quantity could not be updated.
Remove	Removes cart item.	Item could not be removed.
Coupon input and Apply	Uppercases the entered coupon and previews discount.	Enter a coupon code, coupon max 40 characters, coupon applied, coupon not applicable, coupon could not be checked.
Proceed to checkout	Opens Checkout. Disabled when cart is empty or has out-of-stock items.	Remove out-of-stock items before continuing to checkout.

Control	Behavior	Messages
Move to cart	Moves available wishlist item into cart.	Item moved to cart, or item could not be moved to cart.
Wishlist trash icon	Removes item from wishlist.	Item removed from wishlist, or item could not be removed.

7. Checkout, Payment, Orders, And Account

Checkout Controls

Area	Fields / Actions	Rules
Checkout progress	Address, Review, Payment	Shows the current step and completed steps.
Delivery address	Saved-address radio cards	A delivery address is required. If none exists, the UI says to add one from profile before checkout.
Payment method	Card / Stripe, UPI / Razorpay, or configured providers	Payment method is required and comes from <code>VITE_PAYMENT_PROVIDERS</code> .
Review order	Items, quantity, subtotal, discount, total, coupon label	Uses the active cart and saved coupon code.
Create payment	Submits checkout request	Disabled while submitting, if no address, no items, or out-of-stock items exist.
Payment ready panel	Provider, amount, status, provider mount, Continue to payment	Continue button only appears if backend returns <code>redirect_url</code> , <code>payment_url</code> , or <code>next_action_url</code> .

Payment Result Pages

Route	Behavior	Actions
<code>/checkout/success?order_id=...</code>	Shows Order placed, order status, and total. Clears checkout idempotency key and stored coupon after order loads.	Continue shopping.
<code>/checkout/failure?order_id=...&payment_id=...</code>	Shows Payment failed, order status, total, and retry option.	Retry payment or Back to cart.
Missing order id	Shows "Order reference missing".	Open checkout again to confirm latest order status.

Orders

Page	Available Actions	Rules
Orders list	View order cards, status badge, total, date, item count, Previous/Next, View details.	Page size is 10. Empty state links to Browse products.
Order detail	Back to orders, view total, placed date, payment status, fulfillment status, item list, timeline, delivery address.	Cancel order appears only for statuses <code>created</code> , <code>pending_payment</code> , <code>paid</code> , and <code>packed</code> .

Page	Available Actions	Rules
Cancel order dialog	Reason field, Keep order, Cancel order.	Reason must be at least 5 characters. Backend makes the final cancellation decision.

Profile And Addresses

Page	Fields / Buttons	Validation
Profile	Email read-only, Full name, Phone, Avatar URL, Save profile, Cancel, Retry on load failure.	Full name min 2 characters, avatar must be valid image URL if entered.
Addresses	Add address, Add first address, Edit icon, Delete icon, delete confirmation, Save address, Add address, Cancel, Use as default address.	Name min 2, phone min 8 if entered, address line 1 min 5, city/state/country min 2, postal code min 4.
Notifications	Email updates, SMS alerts, Push notifications, Marketing messages, Save preferences, Cancel, Retry.	Each option is a boolean checkbox. Transactional updates are described separately from marketing messages.

Logout

- 1 Open the Account menu.
- 2 Select Logout.
- 3 The app calls `/api/v1/auth/logout`, clears session storage, clears auth state, and removes cached auth/cart/wishlist/profile/order queries.

8. Errors And Troubleshooting

Common User-Facing Messages

Message / Symptom	Likely Cause	What To Do
Checking session...	Protected route is validating the current user.	Wait. If it fails, login again.
Login page opens after Cart/Wishlist/Account	User is not authenticated or session expired.	Login, then retry the protected page.
Network error. Please check your connection and try again.	Frontend cannot reach API Gateway or browser network failed.	Check <code>VITE_API_BASE_URL</code> and Gateway health.
Products could not be loaded / Search results could not be loaded	Product Service, Search Service, Typesense, or API Gateway issue.	Check <code>http://localhost:8080/health/ready</code> , Product Service, Search Service, and Typesense health.
No products available yet	Product API returned no published products.	Publish products from the seller/catalog side or seed local product data.
No saved delivery address	Checkout requires a saved address.	Go to Account -> Addresses and add one.
Remove out-of-stock items before continuing to checkout	Cart contains item with <code>out_of_stock</code> stock status.	Remove the item before checkout.
Payment could not be started	Payment provider response did not include client secret or redirect URL.	Retry checkout and check Payment Service configuration/logs.
Order reference missing	Payment result page opened without <code>order_id</code> .	Open checkout or orders again to verify status.
Phone OTP delivery is not enabled in the local stack	Local SMS provider is disabled.	Use email OTP and Mailpit for local testing.

Useful Local Checks

Frontend	<code>http://localhost:3000/healthz</code>
Gateway ready	<code>http://localhost:8080/health/ready</code>
Products	<code>http://localhost:8080/api/v1/products?page=1&page_size=12&status=published</code>
Categories	<code>http://localhost:8080/api/v1/categories</code>

Autocomplete	<code>http://localhost:8080/api/v1/search/autocomplete?q=sh&limit=3</code>
Mailpit	<code>http://localhost:8025</code>

9. Limitations And Future Improvements

Current Limitations

Limitation	Impact
No seeded buyer/test user found	Logged-in screenshots and end-to-end cart/checkout/order verification could not be completed without creating new local data.
No published products returned by local Product Service	Product listing/detail screenshots with real products could not be captured.
Payment provider SDK UI is partial	The frontend displays a mount placeholder and redirect button when available, but does not embed Stripe/Razorpay SDK forms.
Notification inbox is not present	Users can manage preferences but cannot view an in-app notification list.
Recommendation hooks are not connected to visible pages	No user-facing recommendation carousel or section appears in current routes.
Variant attribute separator may render as <code>Â·</code>	Some variant attribute labels may show an encoding artifact in the UI.

Future Improvements

- Add local buyer seed credentials and safe sample buyer data.
- Seed published products with images and stock for complete browse/cart/checkout screenshots.
- Implement embedded provider SDK forms for Stripe/Razorpay, or clearly force redirect-only payment mode.
- Add in-app notification list if notification viewing is a product requirement.
- Connect recommendation API hooks to visible product/home/cart sections if recommendations are required.
- Add an inline "Add address" action inside checkout to reduce checkout blocking.
- Fix the variant attribute separator encoding artifact.

Final Notes

The manual was prepared from the current User App Frontend source code, route definitions, API clients, runbooks, and live local UI captures. Screenshots are real local UI captures where possible. Authenticated screenshots are marked as placeholders because no seeded buyer/test credentials were available.